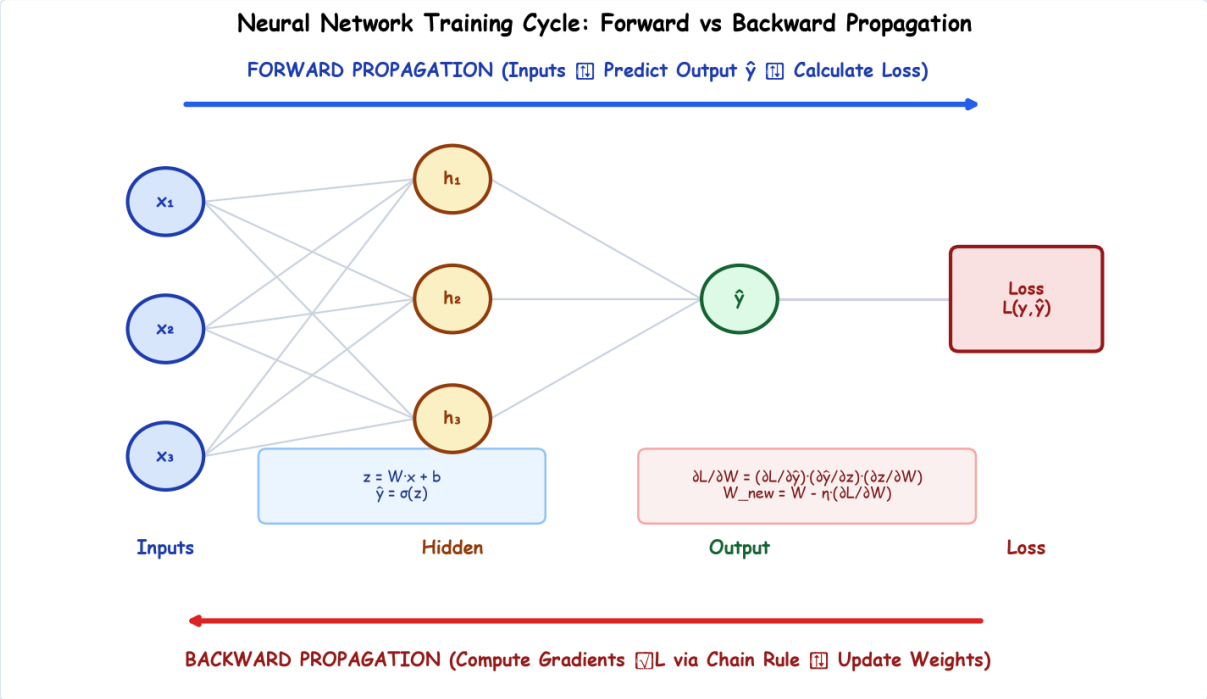
Submodule 37.01: Forward & Backward Propagation (Hinglish)

1. Parichay: Neural Network Training ke Do Engine

Neural Network ko train karne ka process do sequential steps par adharit hota hai:

1. **Forward Propagation (Aage Badhne ka Phase)**: Input features pehle input layer me aate hain, fir hidden layers se guzar kar prediction $\hat{y}$ nikalte hain, aur aakhir me loss function se error evaluate hota hai.
2. **Backward Propagation (Peeche Lautne aur Seekhne ka Phase)**: Error nikalne ke baad, network peeche ki taraf chalta hai aur Calculus ke **Chain Rule** ki madad se har ek weight (W) aur bias (b) ka gradient nikal kar unhe update karta hai.



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2. Forward Propagation ki Working

Forward pass me data hamesha left se right move karta hai:

1. **Linear Combination**:

$$z = W \cdot x + b$$

Yahan har input feature ko uske respective weight se multiply karke bias term joda jata hai.

2. **Activation Function**:

$$a = g(z)$$

Is linear sum par non-linear activation (jaise ReLU ya Sigmoid) apply kiya jata hai taaki model complex patterns seekh sake.

3. **Loss Calculation**: Model ke predicted output $\hat{y}$ aur actual ground truth y ke beech ka loss calculate kiya jata hai.

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3. Backward Propagation ki Working

Jab loss calculate ho jata hai, tab network **Gradient Descent** algorithm ki madad se seekhta hai:

- Network dekhta hai: *"Kis weight ki wajah se kitna error aaya?"*
- Is process ko calculate karne ke liye hum **Partial Derivatives** (Gradients) nikalte hain: $\frac{\partial \mathcal{L}}{\partial W}$ aur $\frac{\partial \mathcal{L}}{\partial b}$.
- Gradient nikalne ke baad naye weights update hote hain:

$$W_{new} = W_{old} - \eta \cdot \frac{\partial \mathcal{L}}{\partial W}$$

$$b_{new} = b_{old} - \eta \cdot \frac{\partial \mathcal{L}}{\partial b}$$

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4. Forward vs Backward Pass ka Tulnatmak Chart

Feature	Forward Propagation	Backward Propagation
Direction	Input $\rightarrow$ Hidden $\rightarrow$ Output	Output $\rightarrow$ Hidden $\rightarrow$ Input
Main Kaam	Output $\hat{y}$ aur Loss $\mathcal{L}$ nikalna	Gradients calculate karna aur Weights update karna
Math Tool	Matrix multiplication + Activation	Chain rule of derivatives

| **Kab Chalta Hai?** | Training aur Testing (Deployment) dono me | Sirf aur sirf Training phase me |\n